Here is the detailed algorithmic extraction and system breakdown for **Batch 3 (Videos 21 through 30)** of the Trade City Pro technical analysis series.
This batch transitions from the initial indicator foundations into the core engines of the system: **Trend Quality & Health Analysis (Trend Analysis / TR)**, **Candle Sizing Dynamics**, **Entry Filters (Price vs. ATR Volatility)**, **Trendline Construction Geometry**, and **The Volume & Open Interest Foundations**.

Video 21: "شکست و تثبیت حمایت و مقاومت بر اساس پرایس اکشن" (Part 21)

- **Core Concept**: Algorithmic rules for authentic breakout validation versus fakeout sweeps, touch counts, and timeframe consistency 1, 2.
- **Algorithmic Rules & Invalidation Logic**:
- **The Structural Body-Close Rule**: A breakout is registered **strictly** when the real body of the candle closes past the horizontal level 2. Wicks that pierce and pull back are classified as liquidity sweeps or false breakouts 2.
- **Timeframe Invalidation Anchor**: A breakout belongs to the timeframe where the level was drawn 2. If a level was mapped on the 4-Hour chart, a 15-minute or 1-hour close above that level has zero breakout validity; the algorithm must wait for the 4-Hour candle close before triggering continuation logic 2.
- **Touch-Count Boundary**:
- *1st Touch*: High failure rate; execution prohibited 2.
- *2nd Touch*: High risk; requires secondary confluence (e.g., higher-timeframe trend continuation or seller exhaustion wick) 2.
- *3rd & 4th Touch*: Optimal institutional breakout window 2. Repetitive testing depletes the passive limit order walls, maximizing continuation probability 3, 4.
- **Effort vs. Result Verification**: A valid breakout candle must display an above-average real body closing near its extreme, accompanied by evidence of opposing rejection (e.g., a long lower wick confirming buyers absorbed sellers during the push) 2.

Video 22: "ترند آنالیز (سلامت روند) - ترند آنالیز پارت ۱" (Part 22)

- **Core Concept**: Quantitative measurement of trend health, velocity, and sustainability through swing amplitudes and cycle timing 5, 6.
- **Algorithmic Mechanics**:
- **Swing Amplitude Degradation (Cycle Amplitude)**:
- In a healthy uptrend, each progressive impulse leg must expand higher than its predecessor 6.
- When the net vertical distance between successive swing highs begins contracting (e.g., Leg 1: +10%, Leg 2: +6%, Leg 3: +2%), the trend's momentum is exhausted 6, 7. The algorithm must halt trend-continuation entries and tighten stops 6, 7.
- **Impulse-to-Correction Asymmetry**:
- For a Bullish Trend: Impulse duration (in candle count) and distance must substantially exceed correction duration and distance (e.g., 15 candles up, 3–5 candles down) 6, 8, 9.
- When the number of candles required to generate a new high doubles or triples relative to previous waves, algorithmic trend weakness is flagged 8, 10.

- **Candle Uniformity & Predictability**:
- **High-Quality Trend**: Formed by uniform, medium-sized candles advancing at an orderly pace 11, 12.
- **Low-Quality / Climax Trend**: Dominated by erratic, massive candles ("whale spikes") 12, 13. These represent late-stage retail FOMO or liquidation sweeps and typically precede immediate, sharp corrections 12, 13.

Video 23: "پرایس اکشن این قسمت ترند آنالیز سایز کندل ها - ترند آنالیز پارت ۲" (Part 23)

- **Core Concept**: Evaluating trend decay and momentum transitions via candlestick geometry (Body-to-Range ratios, Close progression, Curvature, and Channel Excursions) 9, 14-16.
- **Algorithmic Metrics**:
- **Metric A: Real Body-to-Range Ratio (\\(\frac{|\text{Close} - \text{Open}|}{\text{High} - \text{Low}}\\))**:
- In a healthy impulse, \\(\frac{\text{Body}}{\text{Range}} \ge 0.60\text{–}0.70\\) 14.
- When this ratio progressively decays below \\(0.30\\) while price is advancing, buyers are failing to hold intraday gains against limit sellers 14.
- **Metric B: Close Placement Progression**:
- In an uptrend, track where each candle closes within its total range 15.
- If prices are making marginally higher highs, but successive closes occur lower down the candle (e.g., closing at the top 10% \\(\to\\) top 30% \\(\to\\) mid-point \\(\to\\) lower half), sell-side absorption is confirmed 15, 17.
- **Metric C: Trend Curvature (Concave vs. Convex Parabolic)**:
- *Convex Acceleration*: Price advances with a steepening angle; highly volatile, prone to instant collapse 9.
- *Concave Deceleration*: Slope begins flattening out toward a horizontal tangent 9. Programmatic alert for a trend reversal 9.
- **Metric D: Channel Excursion (Price Exhaustion)**:
- When price breaks out through the **overside** of an existing parallel trend channel (e.g., piercing above the upper boundary of an ascending channel), do **not** buy 16. In 80–90% of instances, this is an exhaustion move (*Price Exhaustion*) that rapidly pulls back inside the channel 16.

Video 24: "پرایس اکشن ترند آنالیز انواع روش تایید برای ورود به ترید" (Part 24)

- **Core Concept**: The four formal algorithmic entry confirmation filters: Price-Based, Volatility/Scale-Based (ATR), Time-Based, and Event-Based 18-20.
- **Algorithmic Filter Taxonomy**:
- **Price-Based Filter (The Anchor)**:
- Entry triggers strictly when price closes beyond a discrete numerical threshold (e.g., \\(\text{Close} > \$40,254\\)) 19.
- **Volatility / Scale-Based Filter (ATR Multiplier)**:
- Calculates the average range of the preceding 6 candles 21, 22:
- \\\text{AvgRange}*6 = *frac{1}{6}*\sum*{i=1}^{6} (\text{High}_i - \text{Low}_i)\\

- **Breakout Rule**: The breakout candle's range must be at least **\\(1.5\times\\) the 6-period average range** 21. A small candle that barely crosses a line fails the volatility test and is discarded 21, 23.
- **Time-Based Filter**:
- Requires \\(N\\) consecutive candles to maintain closes beyond the broken boundary 20, 23:
- 15-Minute chart: Requires **8 consecutive candles** 23.
- 4-Hour chart: Requires **4 consecutive candles** 23.
- Daily chart: Requires **2 consecutive candles** 23.
- **Event-Based Filter (Market Structure)**:
- Waits for an objective structural market event (e.g., a formal Higher High breach following a Higher Low formation) 20.
- **Master Entry Recommendation**: Combine **Price-Based** with the **6-period Volatility Ratio (\\(\ge 1.5\times\\))** to avoid getting trapped by sluggish fakeouts 22, 24.

Video 25: "پرایس اکشن روند شناسی و رسم روند و ترند آنالیز" (Part 25)

- **Core Concept**: Rigorous rules for constructing, adjusting, and invalidating trendlines across multi-timeframe charts 25-27.
- **Algorithmic Construction Rules**:
- **The First Swing Pivot Rule**: Never anchor a primary trendline from the initial sharp impulse peak or extreme panic spike 26. Anchor from the second confirmed structural swing high/low, which reflects stabilized institutional participation 26.
- **Two-Touch vs. Three-Touch Validity**:
- *2 Points*: A line connecting two points is purely an **unconfirmed hypothesis** 28, 29.
- *3 Points*: A trendline is officially confirmed and activated for execution only after the **3rd distinct test and rejection** 28, 29.
- **The 200-Candle Window Rule**:
- Evaluate and calibrate the trendline across a rolling window of **200 candles** on the chosen timeframe 27.
- If a trend spans more than 200 bars, step up to a higher timeframe where the entire structure fits into roughly 200 bars (e.g., shifting from 1H to 4H) 27, 30.
- **Angular Constraint (Slope Geometry)**:
- **Optimal Angle**: \\(30^\circ\text{–}45^\circ\\) 30.
- **Valid Range**: Must fall between \\(20^\circ\\) (minimum slope) and \\(60^\circ\\) (maximum slope) 30.
- Any structure \\(<20^\circ\\) is classified as a horizontal rectangle/box; any structure \\(>60^\circ\\) is an unsustainable parabolic blow-off 30.

Video 26: "هرچی از تغییر روند می شناسی بنداز دور" (Part 26)

- **Core Concept**: Objective trend reversal identification, Primary vs. Secondary trends, and the Big Indecision Candle trigger 31-33.
- **System Definitions & Rules**:
- **Primary vs. Secondary Waves**:

- Trades taken in alignment with the Primary Trend target \\(5R\text{–}10R+\\) using trailing stops 33.
- Trades taken during Secondary Waves (corrections or premature counter-trend reversals) must use strict, conservative targets (\\(1:2\\) or \\(1:3\\) max) 33.
- **The Giant Indecision Reversal Trigger**:
- Systematic trend reversals rarely occur smoothly; they are marked by a high-volume **Indecision / Long-Wick Rejection Candle** whose range and volume are visibly larger than the preceding 10 bars 32.
- **Execution**: Invalidation is set at the extreme wick tip 32. Entry triggers once price breaks beyond the opposite side of the indecision candle, provided higher-timeframe weakness was already verified 32, 34.
- **Oscillator Exhaustion Rule**: In a strong trend, an oscillator bouncing out of oversold/overbought does **not** signify a reversal—it typically leads to a flat, choppy range before trend continuation 33, 35.

Video 27: "هرچی از ضعف روند و مولتی تایم فریم می شناسی بنداز دور" (Part 27)

- **Core Concept**: Multi-timeframe momentum cascades, Moving Average slope physics, and structural fatigue 36-38.
- **Algorithmic Rules**:
- **The Multi-Timeframe Velocity Hierarchy**:
- Monthly & Weekly define the **Primary Force Vector** 37. Lower-timeframe counter-trend rallies within a bearish macro trend are programmatic bear-market traps 37.
- A higher-timeframe reversal occurs sequentially: momentum shift on 15m \\(\to\\) 1H \\(\to\\) 4H \\(\to\\) Daily 34, 38, 39.
- **Dynamic Slope Velocity Engine (7 SMA & 25 SMA)**:
- Calculate the rate of change of the moving average angle:
- \\\Delta \theta_{\text{SMA}} = \text{Slope}(\text{SMA}_7, t) - \text{Slope}(\text{SMA}_7, t-1)\\
- When \\(\text{Slope}(\text{SMA}_7)\\) transitions from negative toward **0 (horizontal flatline)**, trend velocity has stopped 38, 40.
- When \\(\text{SMA}*7*\\) *and* *(\text{SMA}*{25}\\) repeatedly intersect without directional separation, trigger a **Choppy Regime State** and disable breakout orders 41, 42.
- **Candle Color Ratio Filter**:
- In a healthy uptrend, green candles dominate count and volume 39.
- When the ratio of opposite-colored candles (red bars in an uptrend) increases over a rolling 20-bar lookback, sell-side market orders are entering the order stream 39.

Video 28: "سریع ترین روش تشخیص تغییر روند ارزدیجیتال" (Part 28)

- **Core Concept**: Structural Failure-to-Advance (Dow Theory Breakdowns) and liquidity sweep mechanics 43.
- **Algorithmic Mechanics**:
- **The Failed Peak / Trough Invalidation**:

- A bullish trend officially terminates not when a line breaks, but when price **fails to make a new Higher High** after pulling back to a key level 43, 44.
- **Trigger Condition**: Price forms an equal or lower high, rolls over, and breaks below the neckline (the intermediate low between the two peaks) 43, 44.
- **The Liquidity Absorption Footprint**:
- When an aggressive impulse leg reaches a major support level and stalls without bouncing immediately, limit buyers are absent and passive bids are being consumed by taker sell volume 43.
- A subsequent consolidation hugging the support floor indicates that liquidity below the level is about to be cleared via a liquidation cascade 43.

Video 29: "هرچی از حجم و اوپن اینترست می شناسی بنداز دور" (Part 29)

- **Core Concept**: Volume-Price Action Analysis (VPA), Volume Divergence matrices, and Volume behavior inside Consolidation Boxes 45-47.
- **The 4-State Quantitative Volume Matrix**:
- **Bullish Confirmation**: Price Rises (\\(\uparrow\\)) + Volume Expands (\\(\uparrow\\)). Authentic aggressive buying pressure 47.
- **Bearish Divergence**: Price Rises (\\(\uparrow\\)) + Volume Contracts (\\(\downarrow\\)). Buying momentum is drying up; retail participants are pushing prices into institutional supply 47, 48.
- **Bearish Confirmation**: Price Falls (\\(\downarrow\\)) + Volume Expands (\\(\uparrow\\)). Dominant aggressive market selling 47.
- **Bullish Divergence**: Price Falls (\\(\downarrow\\)) + Volume Contracts (\\(\downarrow\\)). Selling pressure is exhausted; institutional absorption is underway 47, 49.
- **Consolidation Box Volume Rule**:
- Within a horizontal consolidation box, **volume drying up is NOT trend weakness**—it indicates volatility compression and inventory accumulation 46.
- **Action**: Prepare for an explosive breakout 46. A sharp volume surge on a breakout bar confirms execution 50.

Video 30: "انواع اندیکاتورهای حجم و تحلیل زمانی در ارزدیجیتال" (Part 30)

- **Core Concept**: Volume extremes (Buying/Selling Climaxes vs. Low Volume Reversals) and timeframe sensitivity of volume data 51-53.
- **Algorithmic Rules**:
- **Low Volume Reversal Setup**:
- When price tests an established macro resistance or support boundary with volume contracting to cyclical multi-bar lows, institutions are unwilling to commit capital to continuation 51, 54.
- **Action**: Do not enter on the volume dip alone; position a trigger order at the opposite side of the reversal candle to enter on confirmed displacement 51, 54.
- **Climax / Blow-Off Volume (Buying/Selling Climax)**:

- Marked by a massive vertical volume spike (\\(\ge 3\times\text{ to }5\times\\) standard baseline) coinciding with an outsized range bar 52, 55.
- Signifies mass liquidations and retail capitulation 55, 56. This marks an immediate cycle top/bottom, demanding prompt profit taking on open trades 55, 57.
- **Timeframe Inversion for Volume**:
- Unlike classical price action (where higher timeframes are more stable), **volume data is far more sensitive and granular on lower timeframes (15-Minute / 1-Hour)**, where institutional order execution patterns and absorption footprints are directly visible 53, 58.

Suggested Next Step

We are set to proceed with **Batch 4 (Videos 31 through 40)**, which covers:

- Volume-by-Price Profiling & Anchored Volume Zones (Parts 31–33)
- The 3-SMA System: 7, 25, 99 SMAs (Part 34)
- The Proprietary 15-Minute "Candle Setup" Blueprint & Order Types (Parts 35–37)
- The Engulfing Candle Engine (Part 38)
- Order Book Tape Reading & Liquidity Depth Analysis (Parts 39–40)

Let me know if you would like me to generate Batch 4 now.